

Lecture 3: Simple Linear Regression and the Conditional Approach

Alejandro Melo Ponce

Spring 2023

Contents

1	The Least-Squares Estimator	2
2	The conditional approach	16
3	Variance-Covariance Matrix of the LSE	20
4	Distributional Properties of the LSE	24
5	Misspecification Error:	35

1 The Least-Squares Estimator

Simple Linear Regression:

- One dependent variable
- One explanatory variable.

Pedagogic Purposes:

- Easier to understand the intuition
- We can plot the functional relationship on a 2D graph

Later we will consider the case of multiple explanatory variables $\rightarrow$ Multiple Linear regression

We have cross-sectional data, and we want to use it to estimate the functional relationship between Y and X

- This is called simple regression (Y and a single variable X)
- Theory to make inference on the unknown parameters of the functional relationship, assuming that the conditional expectation is a linear function of the explanatory variable $\rightarrow$ simple linear regression.

The theory we are going to develop in this lecture is called the conditional approach since it makes several assumptions on the conditional moment and distribution of the dependent variable given the explanatory variable. → Assumptions are quite strong to be practical → We will later relax those assumptions by appealing to the conditional approach, based on large-sample theory - asymptotic theory in statistics (what happens when we have a very large number of variables, i.e., observations)

Assumption 1 (random sampling) The data is a random sample of independent and identically distributed vectors $(X_1, Y_1), \dots, (X_n, Y_n)$. The sample size is n . Y is the dependent and X is the explanatory variable. Discussion: We have a population described by the r.v. (X, Y) with joint p.d.f. $f_{x,y}(x, y)$. From this population we pick at random, a sample of n elements/observations: $(X_1, Y_1), \dots, (X_n, Y_n)$: each observation:

- is independent of one another
- all are distributed according $f_{x,y}(x, y)$

independent and identically distributed=i.i.d.

Assumption 2: $E[Y|X] = \beta_0 + \beta_1 X$ i.e., the conditional expectation is linear.
 Alternatively: $Y = \beta_0 + \beta_1 X + \varepsilon$, where ε is a random variable such that $E[\varepsilon|X] = 0$
 Then $E[Y|X] = \beta_0 + \beta_1 X + E[\varepsilon|X] = \beta_0 + \beta_1 X$ Discussion:

- Means that, anything else (other variables) that explains Y, via the relationship, $Y = \beta_0 + \beta_1 X + \varepsilon$, when I condition on X, on average, they don't affect Y, i.e., $E[Y|X] = \beta_0 + \beta_1 X$
- Since $E(Y|X)$ is linear, $E(Y|X)$ =BLP

Example: Y=wage/hour, X=years of education $\Rightarrow$ Can we really think that $E[\text{wage}|\text{education}] = \beta_0 + \beta_1 \text{education}$?

Assumption, although very restrictive, is critical for the theory developed in this lecture.

The Least-Squares Estimator We have data and the functional relationship (Assumption 2)

Using the analogy principle (expectations become averages) we will construct estimators for the parameters of interest.

BLP, we minimized the Mean-squared error

$\updownarrow$ analogy principle

Least-squares estimator, we minimize the mean-squared average

$$\rightarrow MSA(b_0, b_1) = \frac{1}{n} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2$$

Note that "1/n" doesn't play a role in terms of finding the optimum, so we drop it and define:

$$\text{Sum-Squares: } \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2 = SS(b_0, b_1)$$

Least-Squares: $(\hat{\beta}_0, \hat{\beta}_1) = \operatorname{argmin}_{b_0, b_1} SS(b_0, b_1)$

LSE $(\hat{\beta}_0, \hat{\beta}_1)$ is an estimator of the parameters (β_0, β_1) .

To obtain expressions for $(\hat{\beta}_0, \hat{\beta}_1)$, we will analyze the first order conditions.

We will need some results first.

Notation: If we have $X_1, \dots, X_n$, the average (sample) is denoted by $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

Lemma 3.0 If we have $X_1, \dots, X_n$, then:

$$\sum_{i=1}^n (X_i - \bar{X}) = 0$$

Proof: $\sum_{i=1}^n (X_i - \bar{X}) = \sum_{i=1}^n X_i - \sum_{i=1}^n \bar{X} = \sum_{i=1}^n n - n\bar{X} = n\bar{X} - n\bar{X} = 0$

Lemma 3.1. If we denote

$$s_{xy} = \sum (x_i - \bar{X})(y_i - \bar{Y}) \text{ and}$$

$$s_{xx} = \sum (x_i - \bar{X})^2$$

then:

$$(i) \ s_{xy} = \sum (x_i - \bar{x})y_i = \sum x_i(Y_i - \bar{Y}) = \sum X_i Y_i - n\bar{X}\bar{Y}$$

$$(ii) \ s_{xx} = \sum (x_i - \bar{x})x_i = \sum x_i^2 - n\bar{X}^2$$

Proof

(i)

$$\begin{aligned} \sum (X_i - \bar{X})(Y_i - \bar{Y}) &= \sum [(X_i - \bar{X})Y_i - (X_i - \bar{X})\bar{Y}] \\ &= \sum (X_i - \bar{X})Y_i - \sum (X_i - \bar{X})\bar{Y} \\ &= \sum (X_i - \bar{X})Y_i - \bar{Y} \sum (X_i - \bar{X}) \\ &= \sum (X_i - \bar{X})Y_i \end{aligned}$$

$$\begin{aligned} \text{Also: } \sum (X_i - \bar{X})(Y_i - \bar{Y}) &= \sum [X_i Y_i - X_i \bar{Y} - \bar{X} Y_i + \bar{X} \bar{Y}] \\ &= \sum X_i Y_i - \sum X_i \bar{Y} - \sum \bar{X} Y_i + \sum \bar{X} \bar{Y} \\ &= \sum X_i Y_i - \bar{Y} \sum X_i - \bar{X} \sum Y_i + n\bar{X}\bar{Y} \\ &= \sum X_i Y_i - \bar{Y} n\bar{X} - \bar{X} n\bar{Y} + n\bar{X}\bar{Y} \\ &= \sum X_i Y_i + n\bar{X}\bar{Y} \end{aligned}$$

(ii) homework

The Least-Squares Estimator

$$\min_{b_0, b_1} SS(b_0, b_1) = \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2$$

↓

$LSE(\hat{\beta}_0, \hat{\beta}_1)$: estimates of the parameters $(\beta_0, \beta_1) \rightarrow E(Y|X) = \beta_0 + \beta_1 X$

- Mathematical Side: What does it mean to minimize the sum of squares?
- Statistical Theory: What are the distributional properties of the LSE

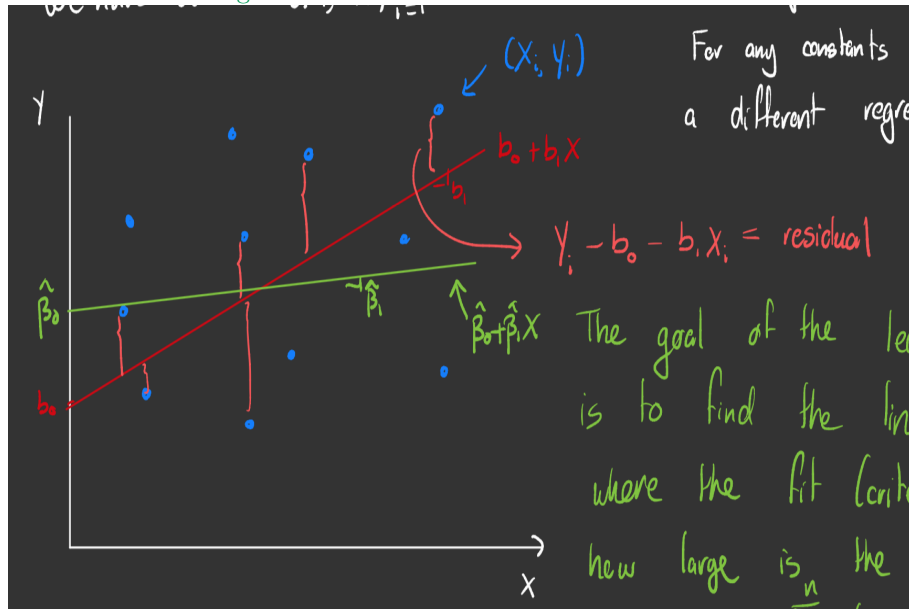
Graphical Interpretation

We have data $(X_i, Y_i)_{i=1}^n$

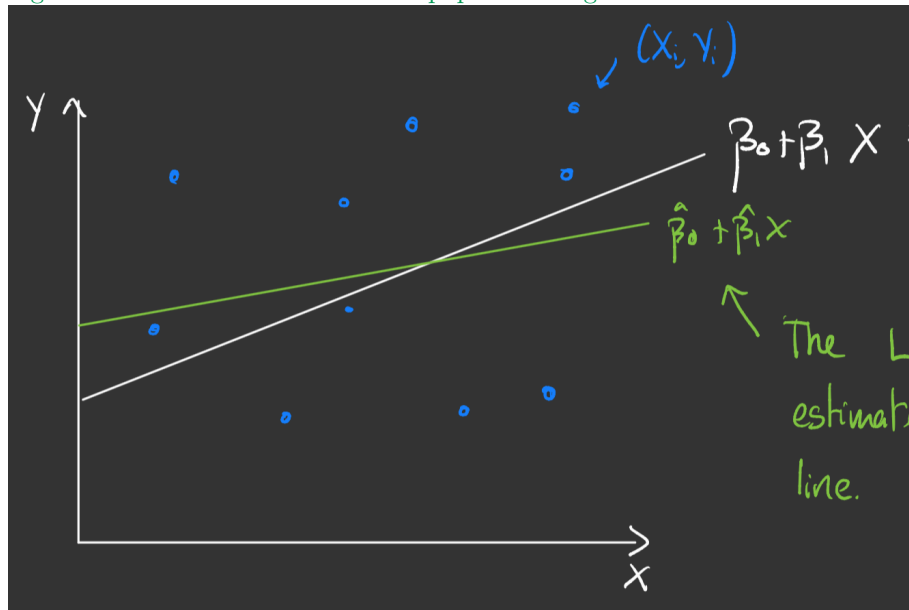
The regression line: $b_0 + b_1x$. For any constants (b_0, b_1) , we have a different regression line.

The goal of the least-squares estimator (LSE) is to find the line of best fit where the fit (criteria) is given by how large is the sum of squared residuals $SS(b_0, b_1) = \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2$

The line of best fit is obtained: $(\hat{\beta}_0, \hat{\beta}_1) = \min_{(b_0, b_1)} SS(b_0, b_1)$: the line $\hat{\beta}_0 + \hat{\beta}_1 X$ is called the LSE regression line.



Remember, the goal is to estimate the unknown parameters β_0, β_1 from the functional relationship $E(Y|X) = \beta_0 + \beta_1 X$. The line $E(Y|X) = \beta_0 + \beta_1 X$ is called the population regression line. Since β_0, β_1 are unknown parameters, we don't know this "line". The LSE regression line aims to estimate the population regression line.



Theorem 3.2 The LSE exists and is unique if and only if $s_{xx} > 0$ and in such case:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \text{ and } \hat{\beta}_1 = \frac{s_{xy}}{s_{xx}}$$

Notice the similarity between the formulas given in Theorem 2.7. and the ones above!

- An estimator of the variance of X : $\widehat{Var(X)} = \frac{1}{n} s_{xx}$
- An estimator of the covariance between (X, Y) : $\widehat{Cov(X, Y)} = \frac{1}{n} s_{xy}$

$$\Rightarrow \hat{\beta}_1 = \frac{s_{xy}}{s_{xx}} = \frac{\frac{1}{n} s_{xy}}{\frac{1}{n} s_{xx}} = \frac{\widehat{Cov(X, Y)}}{\widehat{Var(X)}};$$

$$\text{Similarly } \widehat{E(Y)} = \bar{Y}$$

$$\widehat{E(X)} = \bar{X}$$

Recall, from Theorem 2.7. $\beta_1 = \frac{Cov(X, Y)}{Var(X)}, \beta_0 = E(Y) - \beta_1 E(X)$
Theorem 3.2. is the sample analogue of Theorem 2.7.

Proof: $\min_{b_0, b_1} SS(b_0, b_1) = \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2$
The FOC's from the above problem are: $\frac{\partial SS}{\partial b_0}(\hat{\beta}_0, \hat{\beta}_1) = 0$; $\frac{\partial SS}{\partial b_1}(\hat{\beta}_0, \hat{\beta}_1) = 0$
The above equations lead to:

$$\begin{aligned}\frac{\partial SS}{\partial b_0}(b_0, b_1) &= \sum \left[\frac{\partial}{\partial b_0} (Y_i - b_0 - b_1 X_i)^2 \right] = -2 \sum [Y_i - b_0 - b_1 X_i] \\ \frac{\partial SS}{\partial b_1}(b_0, b_1) &= \sum \left[\frac{\partial}{\partial b_1} (Y_i - b_0 - b_1 X_i)^2 \right] = -2 \sum [(Y_i - b_0 - b_1 X_i) X_i]\end{aligned}$$

Then FOC's are:

$$\begin{aligned}\sum [Y_i - b_0 - b_1 X_i] &= 0 \\ \sum [(Y_i - b_0 - b_1 X_i) X_i] &= 0\end{aligned}$$

which are equivalent to

$$\begin{aligned}\sum Y_i - n\hat{\beta}_0 - \hat{\beta}_0 - \hat{\beta}_1 \sum X_i &= 0 \quad \Longleftrightarrow n\hat{\beta}_0 + (\sum X_i)\hat{\beta}_1 = \sum Y_i \\ \sum Y_i X_i - \hat{\beta}_0 \sum X_i - \hat{\beta}_1 \sum X_i^2 &= 0 \quad \Longleftrightarrow (\sum X_i)\hat{\beta}_0 + (\sum X_i^2)\hat{\beta}_1 = \sum X_i Y_i\end{aligned}$$

The above equations on matrix form are:

$$\begin{bmatrix} n & \sum X_i \\ \sum X_i & \sum X_i^2 \end{bmatrix} \times \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = \begin{pmatrix} \sum Y_i \\ \sum X_i Y_i \end{pmatrix}$$

The determinant of the coefficient matrix is equal to $n(\sum X_i^2) - (\sum X_i)^2$, which is from Lemma 3.1v(ii) is equal to ns_{xx} . This system gives a unique solution (and also gives existence solution) if and only iff $s_{xx} > 0$. If this is the case it follows that

$$\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = \begin{bmatrix} n & \sum X_i \\ \sum X_i & \sum X_i^2 \end{bmatrix}^{-1} \times \begin{pmatrix} \sum Y_i \\ \sum X_i Y_i \end{pmatrix}$$

Homework: Show that the above leads to: $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$; $\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}}$

In here, let's go back:

$$\begin{aligned}\sum Y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum X_i &= 0 \\ \sum X_i Y_i - \hat{\beta}_0 \sum X_i - \hat{\beta}_1 \sum X_i^2 &= 0\end{aligned}$$

From the first equation divide by n to get $\bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X} = 0$
 $\therefore \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$, substitute this into the second equation to get:

$$\begin{aligned}\sum X_i Y_i - (\bar{Y} - \hat{\beta}_1 \bar{X})(\sum X_i) - \hat{\beta}_1 \sum X_i^2 &= 0 \\ \sum X_i Y_i - \bar{Y}(\sum X_i) + \hat{\beta}_1 \bar{X}(\sum X_i) - \hat{\beta}_1 \sum X_i^2 &= 0 \\ \sum X_i Y_i - n\bar{Y}\bar{X} &= \hat{\beta}_1 [\sum X_i^2 - n\bar{X}^2] \\ \Rightarrow \hat{\beta}_1 &= \frac{\sum X_i Y_i - n\bar{X}\bar{Y}}{\sum X_i^2 - n\bar{X}^2} = \frac{S_{XY}}{S_{XX}}\end{aligned}$$

■

Notice that the condition for the existence of the LSE requires $S_{XX} > 0$, in terms of the data is equivalent to the fact that at least two values of the explanatory variable are different.

Assumption 3 $S_{XX} > 0$, non-singularity assumption.

2 The conditional approach

Goal: to get the $\left\{ \begin{array}{c} \text{expected value} \\ \text{other moments} \\ \text{statistical properties} \end{array} \right\}$ of $\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}$, the OLS estimator.

- Recall, by Theorem 3.2., $\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$
- Remember we have a random sample $(X_1, Y_1), \dots, (X_n, Y_n)$
- Since both the X's and Y's are random, S_{XY} and S_{XX} are random variables also, because they are functions of random variables
- Since X and Y are not independent $\Rightarrow S_{XY}$ and S_{XX} are not necessarily independent, and in applications their distribution is not known.
- In order to derive the expected value of $\hat{\beta}$, we need to make some assumptions that will simplify this issue. Since we have assumed that

$$E[Y|X] = \beta_0 + \beta_1 X$$

it makes sense to discuss the statistical properties of the LSE conditional on the values of the observed explanatory variable X.

To keep notation simple, the pdf of a r.v. $(Z_1, \dots, Z_n)$

$$f_{z_1, \dots, z_n}(Z_1, \dots, Z_n) = f(Z_1, \dots, Z_n)$$

The following argument is true for arbitrary n (number of random variables), but I'm going to explain it for $n=2$ for simplicity.
 From assumption 1: (X_1, Y_1) and (X_2, Y_2) are i.i.d. (independent and identically distributed), hence:

$$f(x_1, x_2, y_1, y_2) = f(x_1, y_1)f(x_2, y_2)$$

Recall: Independence between Z_1 and $Z_2 \Rightarrow f_{Z_1}(z_1) = f_{Z_1}(z_1)f_{Z_2}(z_2)$

Identically distributed: $f_{Z_1}(z_1) = f_{Z_2}(z_2)$

Since we are going to condition on $X_1 = x_1, X_2 = x_2$, we compute $f(y_1, y_2|x_1, x_2)$ by

$$f(y_1, y_2|x_1, x_2) = \frac{f(x_1, x_2, y_1, y_2)}{f(x_1, x_2)} = \frac{f(x_1, y_1)}{f(x_2, y_2)} = f(y_1|x_1)f(y_2|x_2)$$

Denominator: $f(x_1, x_2) = \iint f(x_1, y_1)f(x_2, y_2) dy_1 dy_2 = \int f(x_1, y_1) \left(\int f(x_2, y_2) dy_2 \right) dy_1 = \int f(x_1, y_1)f(x_2) dy_1 = f(x_1)f(x_2)$

The result in the box generalizes to n:

$$f(y_1, \dots, y_n | x_1, \dots, x_n) = f(y_1 | x_1) \dots f(y_n | x_n)$$

Homework: Show that $f(y_i | x_1, \dots, x_n) = f(y_i | x_i), i = 1, \dots, n$

This shows that the conditional distribution of Y_i given all the values of the explanatory variable is the same as conditionally on its own value of the explanatory variable.

This property is called strong exogeneity

Notation: We will condition very often on the event $(X_1, \dots, X_n)$, i.e. $(X_1 = x_1, \dots, X_n = x_n)$. We will denote these events as just X's.

Strong exogeneity can be written: $f(y_i | X's) = f(y_i | x_i), i = 1, \dots, n$

Using all the previous observations:

$$\text{From Assumption 2: } E[Y_i|X's] = E[Y_i|X_i] = \beta_0 + \beta_1 X_i, i = 1, \dots, n$$

The properties of the LSE are derived conditioning on the observed values of the X's, thus those values behave as constants in conditional expectations.

3 Variance-Covariance Matrix of the LSE

Assumption 4: $Var(Y|X) = \sigma^2$, i.e. the conditional variance is constant and doesn't vary with X. Homoskedasticity – same variability

This assumption is quite strong and is seldom met in practice.

We focus on this assumption because:

1. Pedagogical Purposes
2. All statistical software gives initial reports on the estimation of the LSE assuming assumption 4.

A very simple example where homoskedasticity doesn't hold is the example from Lecture 1, in that example $Var(Y|X = x) = h(x) \rightarrow$ when $Var(Y-X)$ varies with x, i.e., is not constant, this case is called heteroskedasticity

Recall:
LSE: $\hat{\vec{\beta}} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

$$\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}}$$

A1: $(X_1, Y_1), \dots, (X_n, Y_n)$ are i.i.d.

A2: $E(Y|X) = \beta_0 + \beta_1 X$

A3: $S_{XX} > 0$

A4: $Var(Y|X) = \sigma^2$

Theorem 3.3. Under A1, A2, A3 & A4, we have that

(i) $E(\hat{\vec{\beta}}|X's) = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$

(ii) $Var(\hat{\vec{\beta}}|X's) =$

$$\frac{\sigma^2}{S_{XX}} \times \begin{bmatrix} \frac{1}{n} \sum_{i=1}^n X_i^2 & -\bar{X} \\ -\bar{X} & 1 \end{bmatrix}$$

$\Rightarrow$

$$Var(\hat{\beta}_0|X's) = \frac{\sigma^2}{S_{XX}(\frac{1}{n} \sum X_i^2)}$$

$$Var(\hat{\beta}_1|X's) = \frac{\sigma^2}{S_{XX}}$$

$$Cov(\hat{\beta}_0, \hat{\beta}_1) = \frac{\sigma^2}{S_{XX}}(-\bar{X}) = -\frac{\sigma^2 \bar{X}}{S_{XX}}$$

Proof: (i) Consider $\hat{\beta}_1$ first.

$$\begin{aligned}
E[\beta_1|\hat{X}'s] &= E\left[\frac{S_{XY}}{S_{XX}}|X's\right] \\
&= \frac{1}{S_{XX}}E[S_{XY}|X's] \\
&= \frac{1}{S_{XX}}E\left[\sum(X_i - \bar{X})Y_i|X's\right] \\
&= \frac{1}{S_{XX}}\sum(X_i - \bar{X})E[Y_i|X's] \\
&= \frac{1}{S_{XX}}\sum(X_i - \bar{X})E[Y_i|X_i] \\
&= \frac{1}{S_{XX}}\sum(X_i - \bar{X})(\beta_0 + \beta_1 X_i) \\
&= \frac{1}{S_{XX}}[\beta_0 \sum(X_i - \bar{X}) + \beta_1 \sum(X_i - \bar{X})X_i]
\end{aligned}$$

Recall

Lemma 0 that: $\sum(X_i - \bar{X}) = 0$

Lemma 1.ii that: $\sum(X_i - \bar{X})X_i = S_{XX}$

$$= \frac{1}{S_{XX}}[\beta_0(0) + \beta_1 S_{XX}] = \beta_1 \therefore E[\hat{\beta}_1|X's] = \beta_1$$

The proof for $E[\hat{\beta}_0|X's] = \beta_0$ is left as homework.

I now proceed to derive the expression for $Var(\hat{\beta}_1|X's)$

$$Var(\hat{\beta}_1|X's) = Var\left(\frac{S_{XY}}{S_{XX}}|X's\right) = \frac{1}{S_{XX}^2}Var(S_{XY}|X's) = \frac{1}{S_{XX}^2}Var\sum(X_i - \bar{X})Y_i|X's \Rightarrow$$

I leave as homework to show that if the covariance exists: $Cov(Y_i, Y_j|X's) =$

0, for $1 \leq i \neq j \leq n$

$$\begin{aligned}
Var(S_{XY}|X's) &= \sum Var[(X_i - \bar{X})Y_i|X's] \\
&= \sum (X_i - \bar{X})^2 Var(Y_i|X's) \\
&= \sum (X_i - \bar{X})^2 Var(Y_i|X_i) \\
&\quad \uparrow \text{homoskedasticity \& strong exogeneity} \\
&= \sum (X_i - \bar{X})^2 \sigma^2 \\
&= \sigma^2 S_{XX}
\end{aligned}$$

$$\therefore Var(\hat{\beta}_1|X's) = \frac{1}{S_{XX}^2} Var(S_{XY}|X's) = \frac{1}{S_{XX}^2} \sigma^2 S_{XX} = \frac{\sigma^2}{S_{XX}}$$

The rest of the entries: $Var(\hat{\beta}_0|X's)$, $Cov(\hat{\beta}_0, \hat{\beta}_1|X's)$ are left for you to derive as homework. ■

4 Distributional Properties of the LSE

Assumption 5: The conditional distribution of $Y|X$ is normal, that is, a normality assumption
In practical terms, this implies that, using assumption 2, $E[Y|X] = \beta_0 + \beta_1 X$,
Using Assumption 4, $Var(Y|X) = \sigma^2$
Together with assumptions 1 and 3, we get that

$$Y_i|X_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2)$$

$\Rightarrow Y_i|X_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2) \Rightarrow$ all of those conditional distributions are independent of each other for $i = 1, \dots, n$

To apply the previous assumption, note that (using Lemma 3.1)
 $\hat{\beta}_1$ can be written as a linear combination of the Y 's where the coefficients are only a function of the X 's:

$$\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}} = \frac{1}{S_{XX}} \sum (X_i - \bar{X})Y_i = \sum \frac{(X_i - \bar{X})}{S_{XX}} Y_i$$

$$\therefore \hat{\beta}_1 = \sum C_i(X's) Y_i \text{ where } C_i(X's) = \frac{(X_i - \bar{X})}{S_{XX}}, i = 1, \dots, n$$

$$\Rightarrow \hat{\beta}_1 | X's \sim N(\beta_1, \frac{\sigma^2}{S_{XX}})$$

Homework: verify that $\hat{\beta}_0$ is also normally distributed

Reminder: A linear combination $W = \alpha_1 Z_1 + \dots + \alpha_n Z_n$ of $(Z_1, \dots, Z_n)$ of independent and normally distributed r.v. is also normal.

Theorem 3.4. Under assumptions 1 to 5:

$$\hat{\beta}|X's = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} | X's \sim N_2 \left[\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}, \frac{\sigma^2}{S_{XX}} \begin{pmatrix} \frac{1}{n} \sum_{i=1}^n X_i^2 & -\bar{X} \\ -\bar{X} & 1 \end{pmatrix} \right]$$

Note that in the expression above, only σ^2 has not been estimated! Recall that $\sigma^2 = Var(Y|X)$ under assumption 4. We need a new definition.

Definition 3.1. Consider the dataset $(X_1, Y_1), \dots, (X_n, Y_n)$ such that the LSE exists and is unique. Let $\hat{\beta}$ be the LSE β . We define the fitted model by

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$

the fitted values are: $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i; i = 1, \dots, n$

and residuals are: $\hat{\varepsilon}_i = Y_i - \hat{Y}_i; i = 1, \dots, n$

$\hat{Y}$ estimates $E[Y|X]$ in general:

$\hat{Y}_i$ estimates $E[Y|X_i]$ for the specific value X_i

$\hat{\varepsilon}_i$ estimates the error made by the predictor on the i-th observation

→ Recall from Thm 2.6. $\varepsilon = Y - E(Y|X) \Rightarrow Var(\varepsilon|X) = Var(Y|X) = \sigma^2$

It may be shown that, for simple linear regression, a conditionally unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum (\hat{\varepsilon}_i)^2$$

Conditionally unbiased means that: $E[\hat{\sigma}^2|X] = \sigma^2$

We can now estimate the conditional variance-covariance matrix of $\hat{\beta}$ as

$$\hat{V}(\hat{\beta}|X's) = \frac{\hat{\sigma}^2}{S_{XX}} \begin{bmatrix} \frac{1}{n} \sum X_i^2 & -\bar{X} \\ -\bar{X} & 1 \end{bmatrix}$$

The terms in the diagonal are $\hat{Var}(\hat{\beta}_i|X's)$ $i = 1, 2$ The **standard error of the estimator** is: Std. Error $(\hat{\beta}_i|X's) = SE(\hat{\beta}_i|X's) = \sqrt{\hat{Var}(\hat{\beta}_i|X's)}$

Fact:

$$\frac{(n-2)}{\sigma^2} \hat{\sigma}^2 | X's \sim \chi^2(n-2)$$

↑ chi-squared distribution with n-2 degrees of freedom.

Fact: $\vec{\hat{\beta}}$ and $\hat{\sigma}^2$ are independent → This will allow us to construct hypothesis tests and confidence intervals.

$$\text{t-statistic} := t = \frac{\hat{\beta}_i - \beta_i}{SE(\hat{\beta}_i | X's)} | X's \sim \frac{t(n-2)}{\text{student's t-dist with n-2 degrees of freedom}}$$

"Pivot"

Pivotal quantity: Is a random variable whose distribution does not depend on the parameters that need to be estimated.

Example $t = \frac{\hat{\beta}_i - \beta_i}{SE(\hat{\beta}_i | X's)} | X's \sim t(n-2)$

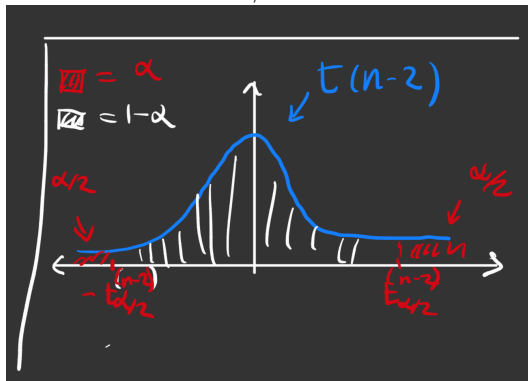
↑ This distribution doesn't depend on neither β nor σ^2

$\therefore t$ is a pivotal quantity

$\Rightarrow$ A confidence interval with $(1 - \alpha)\%$ confidence level is obtained as

$$\left(\hat{\beta}_i - t_{\alpha/2}^{(n-2)} SE(\hat{\beta}_i | X's), \hat{\beta}_i + t_{\alpha/2}^{(n-2)} SE(\hat{\beta}_i | X's) \right)$$

with $Pr(|t_{(n-2)}| \leq t_{\alpha/2}^{(n-2)}) = 1 - \alpha$



To test the hypothesis

$$H_0 : \beta_i = \beta_i^{hyp} \text{ vs } H_1 : \beta_i \neq \beta_i^{hyp}$$

we use as test statistic

$$t = \frac{\hat{\beta}_i - \beta_i}{SE(\hat{\beta}_i|X's)}|X's, \text{ under the null, this quantity is } t(n-2)$$

For an $\alpha\%$ significance level, the critical region is

$$R = \left\{ |t_{calc}| > t_{\alpha/2}^{(n-2)} \right\}$$

with a corresponding P-value of: $Pr(|t(n-2)| > |t_{calc}|)$

Example 3.1

Y =wage=average wage/hour in dollars \$ - dependent variable

X =education=years of formal education - explanatory variable

Research question? What is the ceteris paribus effect of education on wage (effect is in dollars \$)

$n=526$

The results say that:

$$\hat{Y} = -0.90485 + 0.5136X$$

This is an estimate of $E[\text{wage}|\text{education}] \Rightarrow E[\widehat{\text{wage}|\text{education}}] = \widehat{\text{wage}}$

$$\widehat{\text{wage}} = \underset{(0.68497)}{(-0.90485)} + \underset{(0.05325)}{0.5136} \text{ education}$$

estimated equation/regression

To estimate the residuals:

$$\widehat{\text{wage}}_i = -0.90485 + 0.5136 \text{ education}_i; i = 1, \dots, 526$$

$$\hat{\varepsilon}_i = \text{wage}_i - \widehat{\text{wage}}_i; i = 1, \dots, 526$$

$$\widehat{\sigma^2} = 11.41352$$

$$\hat{\sigma} = \sqrt{\widehat{\sigma^2}} = 3.3783$$

$\hat{\sigma}$ = Residual Standard error or standard error of the regression

	Coefficient	Std. Error	t-value ($H_0 : \beta_1 = 0$)	P-value
Intercept $\hat{\beta}_0$	-0.090485	0.68497	-1.321	0.187
education $\hat{\beta}_1$	0.54136	0.05325	10.167	2.77×10^{-22}

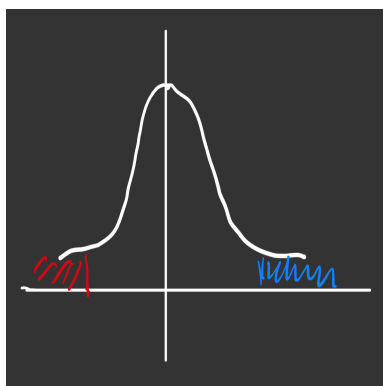
H_0 : education plays no role in the wage H_1 : education plays a role in the wage
VS

$$H_0: \beta_1 = 0$$

$$H_1: \beta_1 \neq 0$$

$$t_{calc} = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1 | X's)} = \frac{0.54136 - 0}{0.05325} = 10.167$$

$$\begin{aligned}
 P\text{-value} &= Pr(|t_{(524)}| > |t_{calc}|) = Pr(|t_{(524)}| > 10.167) = 2.77 \times 10^{-22} \\
 &= Pr(t_{(524)} > 10.167) \cup t_{(524)} < -10.167) \\
 &= Pr(t_{(524)} > 10.167) + Pr(t_{(524)} < -10.167) \\
 &= Pr(t_{(524)} \in (10.167, \infty)) + Pr(t_{(524)} \in (-\infty, -10.167))
 \end{aligned}$$



We want 95% for β_1 :
 $t_{\frac{0.05}{2}} = 1.96$

$$\begin{aligned}(\hat{\beta}_1 \pm t_{0.05/2}^{524} SE(\hat{\beta}_1|X's)) &= (0.541359 \pm 1.96100532) \\ &= (0.04536164, 0.62910)\end{aligned}$$

$\Rightarrow$ Ceteris Paribus effect of education on wage with 95% confidence is between 0.44 and 0.65 dollar/hour per year of education, i.e. 44 cents and 65 cents per year of education

5 Misspecification Error:

Failure of A2: $E[Y|X] = \beta_0 + \beta_1 X$ in the following senses:

- Missing Relevant explanatory variable. X is not the only variable, there is another variable Z that goes into the CEF.
- Irrelevant explanatory variable. A2 is true: $E[Y|X] = \beta_0 + \beta_1 X$, but you include in the regression variable W .

Missing Relevant Explanatory variable

Suppose the true model is $E[Y|X, Z] = \beta_0 + \beta_1 X + \beta_2 Z$ but the researcher fits the model $E^*(Y|X) = \beta_0 + \beta_1 X$, which is the BLP of Y given X , which is different from the conditional expectation.

→ LSE of the slope still is: $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{1}{S_{xx}} \sum_{i=1}^n (X_i - \bar{X})Y_i$

Taking the expectation:

$$\begin{aligned}
E(\hat{\beta}_1|X's, Z's) &= E\left(\frac{1}{S_{xx}} \sum_i (X_i - \bar{X}) Y_i | X's, Z's\right) \\
&= \frac{1}{S_{xx}} \sum_i (X_i - \bar{X}) E(Y_i | X's, Z's) \\
&= \frac{1}{S_{xx}} \sum_i (X_i - \bar{X}) E(Y_i | X_i, Z_i) \\
&= \frac{1}{S_{xx}} \sum_i (X_i - \bar{X}) (\beta_0 + \beta_1 X_i + \beta_2 Z_i) \\
&= \frac{1}{S_{xx}} \left\{ \beta_0 \sum_i (X_i - \bar{X}) + \beta_1 \sum_i (X_i - \bar{X}) X_i + \beta_2 \sum_i (X_i - \bar{X}) Z_i \right\} \\
&= \beta_1 + \beta_2 \left(\frac{S_{xz}}{S_{xx}} \right)
\end{aligned}$$

Since Z is a relevant variable, β_2 has to be different from zero.

S_{xz} : is basically an estimate of the covariance of X and Z . Note, if X and Z were uncorrelated this term would be zero, but it is not generally the case! $\therefore$ The LSE for the slope $\hat{\beta}_1$ is not unbiased (as in Theorem 3.3). The bias is given by:

$$Bias(\hat{\beta}_1|X's, Z's) = \frac{S_{xz}}{S_{xx}} \beta_2$$

Under the rest of the assumptions, the conditional distribution of $\hat{\beta}_1$ is still normal but with mean $\beta_1 + \frac{S_{xz}}{S_{xx}} \beta_2 \neq \beta_1$

$\rightarrow$ anything we construct using the biased estimator $\hat{\beta}_1$ will have the same problem, i.e., confidence intervals and hypothesis tests.

- Suppose that from theoretical estimations that we believe $\beta_1 > 0$, i.e., the true ceteris paribus effect of X on Y is positive; but suppose that the $Bias(\hat{\beta}_1|X's, Z's) = \frac{S_{xz}}{S_{xx}}\beta_2 < 0$.
 - $\hat{\beta}_1$ underestimates the true effect β_1
 - If the $Bias(\hat{\beta}_1|X's, Z's) = \frac{S_{xz}}{S_{xx}}\beta_2$ is negative enough so as to make $E(\hat{\beta}_1|X's, Z's) = \beta_1 + Bias(\hat{\beta}_1|X's, Z's) < 0$ this may provide evidence that the model is incorrectly specified.
- Again suppose that $\beta_1 > 0$, and $Bias(\hat{\beta}_1|X's, Z's) > 0$
 - $\hat{\beta}_1$ overestimates the true effect of β_1 .
 - You should be on the lookout for estimates of $\hat{\beta}_1$ that are "too large" if you believe that you have a missing variable Z.

Sometimes the reason for an incorrectly specified model with a missing variable Z , is that we don't have data on this variable, so there is not much we can do in this case except being careful when using our estimates.

Example 3.2.

Y =wage, X =years of education. There are many omitted variables: years of experience, years of tenure on the job, race, gender and so on.

For example, let Z =years of experience.

The ceteris paribus effects

$$\begin{aligned}\frac{\partial E[Y|X, Z]}{\partial X} &= \beta_1 > 0 \text{(by theoretical considerations)} \\ \frac{\partial E[Y|X, Z]}{\partial Z} &= \beta_2 > 0 \text{(by theoretical considerations, experience)}\end{aligned}$$

$Bias(\hat{\beta}_1|X's, Z's) = \frac{S_{xz}}{S_{xx}}\beta_2 > 0$ since education & experience can be assumed to be positive. $S_{xz} > 0$

$\therefore$ overestimating the effect of education.

Example 3.3.

Y=Savings

X=Income

Z=Expenditure

Suppose that we have the following relation:

$$Savings = 4 + 2Income - 4Expenditure + \varepsilon$$

We are going to do a simulation assuming we know $\beta_0, \beta_1, \beta_2, \sigma^2, \varepsilon$

ε is a r.v. with mean

$$E[\varepsilon|X, Z] = 0$$

$$Var(\varepsilon|X, Z) = \sigma^2$$

$$\varepsilon|X, Z \sim N(0, \sigma^2)$$

Note that $E[Savings|X, Z] = 4 + 2Income - 4Expenditure$

$\Rightarrow \beta_0 = 4, \beta_1 = 2, \beta_2 = -4$

$\therefore \beta_1, \beta_2$ are the true ceteris paribus effects of income and expenditure on savings, i.e.,

$$\frac{\partial E[Savings|Income, Expenditure]}{\partial Income} = \beta_1; \quad \frac{\partial E[Savings|Income, Expenditure]}{\partial Expenditure} = \beta_2$$

An irrelevant explanatory variable is included

In this case A2 $E(Y|X) = \beta_0 + \beta_1 X$, but in your regression you include W, since W is irrelevant.

$$E[Y|X, W] = E[Y|X] = \beta_0 + \beta_1 X$$

$\Rightarrow$ the coefficient on W, say $\beta_2 = 0$

$\Rightarrow E(\hat{\beta}_1|X, W) = \beta_1 \Rightarrow \hat{\beta}_1$ is conditionally unbiased! This kind of misspecification error is not as critical as the previous one!

For the moment take this as fact when we study multiple linear regression we will be able to compute this.

Discussion:

We have exemplified with simple linear regression the Conditional Approach:
We make strong assumptions on the conditional moments of the distribution of the variables of interest.

→ We use those assumptions to derive the statistical properties of the LSE.

→ Although this approach can be extended to multiple explanatory variables, its assumptions are still too strong.

!they may hardly be satisfied by
real world data!

Because of this we will develop the asymptotic approach for multiple linear regression, which is based on large sample theory.

In terms of the assumptions

A1 is given from the data collection procedure. It provides independence of the random vectors that contain the dependent and explanatory variables.

A2 is hard to justify in practice, because we don't know the true distribution of the data. What we are gonna do is to estimate instead the BLP; which works very well with the asymptotic approach.

A3 will also be kept in the asymptotic approach.

A4 (constant conditional variance) is very difficult to observe in practice. **The alternative is heteroskedasticity** $Var(Y|X) = f(X)$ most-real world data exhibits heteroskedasticity.

The asymptotic approach doesn't need A4 (homoskedasticity), we will find an estimator for the variance-covariance matrix of the LSE that is constant under general forms of heteroskedasticity → called the HC estimator of the variance-covariance matrix.

A5 (normality of $Y|X$) This is difficult to sustain. Skewed distributions of Y discrete automatically violate this assumption. The asymptotic approach doesn't require normality.

Goal: The asymptotic approach (based on large-sample theory) is very flexible.
--

→ Next topic: large-sample theory